Scalable Generative Reconstruction of Three-Dimensional Fields on Point Clouds

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Abstract

Reconstructing three-dimensional physical fields from sparse measurements is an ill-posed inverse problem for which generative models provide the natural formulation: they sample from the distribution over fields consistent with the data rather than regressing to its mean. Current methods do not survive the transition to three dimensions—voxel diffusion computes globally on dense grids and hits a measured memory wall, while latent methods compress fields through an autoencoder that bounds what can be recovered. We introduce LOCUS, a conditional rectified-flow model that generates fields directly in ambient function space, on arbitrary point clouds, with no autoencoder and no voxelization. Each query point is conditioned through a compact global summary of the observations and a local, importance-weighted retrieval over sensor tokens; the training objective, an integral over the domain, is estimated on a random ∼2% subset of points per step, decoupling training cost from resolution. A single model is amortized over sensor counts, layouts, noise levels, and observed-channel patterns. On spatially disjoint isotropic-turbulence evaluation at two million points, LOCUS matches or beats a strong latent flow-matching baseline under the CRPS proper score and produces the best-calibrated predictive distributions of any method evaluated, in a literature where every scored precedent is underdispersed. Its training memory is constant from 64^3 to 1024^3 output resolution, where voxel diffusion and the autoencoder hit measured walls at 320^3 and 640^3, and its inference is constant-memory with time linear in the number of evaluation points. Under noisy, occluded, and channel-dropped sensing on wildfire simulation its worst-case error is below every baseline's best case, and it infers volumetric flow from surface-only sensors on unstructured transonic-wing meshes that grid-based generative methods cannot represent.

1 Introduction

Volumetric fields govern systems that experiments can only observe pointwise: velocimetry resolves planes of a turbulent flow while the surrounding volume goes unmeasured, and flight-test campaigns record surface pressure at taps while the off-body flow that determines performance is inaccessible. Recovering the full three-dimensional state is a long-standing inverse problem (Callaham et al., 2019; Raissi et al., 2020; Vinuesa et al., 2023), and at realistic sensor densities it is severely ill-posed: many fields fit the data, and the missing information sits disproportionately at the small scales. Deterministic reconstruction networks (Fukami et al., 2021; Santos et al., 2023; Dang et al., 2025) return a conditional mean—the optimal point estimate and a systematically over-smoothed one, blurring precisely the fine structure whose recovery motivates reconstruction. Generative models resolve the mismatch by sampling fields consistent with the observations, retaining sharp realizations and quantifying what the measurements leave undetermined (Shu et al., 2023; Li et al., 2024; Zhuang et al., 2025).

This posterior-sampling program has largely been carried out in two dimensions (Shu et al., 2023; Amorós-Trepat et al., 2026; Li et al., 2024; Huang et al., 2024; Zhuang et al., 2025). In parallel, generative modeling has been lifted from fixed grids to function space, defining flows over functions that are consistent across discretizations (Kerrigan et al., 2024b; Lim et al., 2025; Pidstrigach et al., 2024; Shi et al., 2025), with guided variants for inverse problems (Yao et al., 2025; Kerrigan et al., 2024a). These methods establish the right abstraction for scattered measurements, but their demonstrations remain at 64×64 to 128×128 on uniform grids with spectral backbones that presuppose them; scaling guided function-space sampling beyond two dimensions is stated as an open problem (Yao et al., 2025).

The transition to three dimensions has been attempted along two detours, each with a documented cost. The first is voxelization: dense 3D U-Nets and transformers compute globally on the full grid at every step, so activation memory grows with the cube of resolution, and published 3D generative models cap their domains at $192 \times 48 \times 48$ (Lienen et al., 2024), train on 32^3 crops with attention removed (Oommen et al., 2026), factorize attention at 64^3 (Li et al., 2023a), or resort to patch decompositions (Holzschuh et al., 2026); it also forfeits the geometry, since unstructured meshes must first be resampled onto axis-aligned grids. The second is latent compression (Du et al., 2024; Zhou et al., 2025; Wang et al., 2026; Li et al., 2025), which makes 3D generation affordable but bounds reconstruction by the autoencoder, and that bound is not tight enough for turbulence: MSE-trained autoencoders retain roughly a quarter of the spectral power in the dissipation range (Rafiq & Nair, 2026; Glaws et al., 2020), and latent neural-field models report small-scale discrepancies on their own benchmarks (Du et al., 2024). A model whose prior cannot represent the small scales cannot return them, whatever the observations. Meanwhile the architectures that do scale to realistic 3D discretizations (Luo et al., 2025; Alkin et al., 2024; 2025; Li et al., 2023b) are uniformly deterministic, and generative modeling on meshes has been limited to a few thousand nodes (Lino et al., 2025).

A further gap separates published evaluations from deployed sensing. The dominant protocol conditions on noiseless values at freely chosen interior locations (Shu et al., 2023; Huang et al., 2024; Amorós-Trepat et al., 2026), but real measurements are noisy, often confined to surfaces (Guastoni et al., 2021; Cuéllar et al., 2024; Mousavi & Eldredge, 2025), occluded by the phenomena being measured, and frequently of different quantities than the ones sought (Barwey et al., 2021; Kildare et al., 2024). That regime is where the generative formulation earns its cost: with noiseless, well-placed observations deterministic models are competitive (Zhuang et al., 2025).

We introduce LOCUS, which performs conditional rectified flow directly on field values at scattered points—ambient space, end to end, no autoencoder and no voxel grid—with a Gaussian-process source that ties the pointwise flow into a coherent flow over functions. Tractability follows from two decisions. First, conditioning is factored by locality: a cross-attention encoder compresses the sensor set into a compact global summary at cost linear in the number of sensors, and each query retrieves a small, importance-weighted neighborhood of sensor tokens at constant cost, so the network never forms a global volumetric representation. Second, the flow-matching objective is a domain integral admitting unbiased estimation on random point subsets, so LOCUS trains on roughly two percent of the discretization per step, decoupling training memory from resolution. Neither decision is available to a voxel U-Net or patchified transformer, whose computation graphs are global in the discretization. Conditioning is amortized over randomized sensor configurations, so one model serves a broad family of counts, placements, noise levels, and observed-channel subsets—including operators beyond the training ranges (Section 4.3)—and rectified flow yields samples in two to sixteen network evaluations.

Our contributions support one central thesis: ambient pointwise conditional generation removes the 3D discretization bottleneck of generative field reconstruction, without latent compression. Specifically:

- Model. The first generative model reconstructing 3D fields at realistic discretization sizes directly in ambient function space, combining locality-factored conditioning, subsampled function-space training, and a binned spectral objective on the rectified-flow endpoint (Section 3).
- Measured scaling separation. Training memory and step time are constant in output discretization from 64^3 to 1024^3 (39.9 GB, 0.44 s) and inference is constant-memory (1.4 GB) with time linear in point count, while voxel diffusion and the latent base-

line's autoencoder hit measured out-of-memory walls at 320^3 and 640^3: grid methods must supervise every cell at every step, whereas LOCUS supervises a chosen 39k-point budget (Section 4.2).

- Accuracy with calibration under honest evaluation. On a spatially disjoint held-out region of forced isotropic turbulence—a protocol adopted after showing the standard random-in-time split reduces to near-duplicate interpolation—LOCUS beats a parameter-matched latent flow-matching baseline on CRPS at one operating point and is TOST-equivalent at the other, concedes at most a small bounded relative-L_2 deficit, and is decisively better calibrated ($t \approx 9$–28 over 50 paired snapshots); a retrained voxel-diffusion baseline trails on both axes (Section 4.1).
- Deployed sensing. Under realistic operators on wildfire LES (noise, slab occlusion, channel dropout), one amortized LOCUS degrades gracefully—its worst case (0.60) beats the best baseline's clean case (0.72)—infers unobserved fields through the same interface, and, on 673 transonic wing meshes, infers volumetric predictive distributions from surface-only sensors, an operator grid-based generative baselines cannot represent (Section 4.3). Throughout we treat the sampled distributions as approximations to the posterior whose calibration we measure rather than assume.

2 Related Work

Sparse field reconstruction. Deterministic learned reconstruction spans convolutional models on Voronoi-embedded sensor images (Fukami et al., 2021), attention over arbitrary sensor sets (Santos et al., 2023; Dang et al., 2025), shallow decoders (Erichson et al., 2020), and wall-to-volume estimation (Guastoni et al., 2021; Cuéllar et al., 2024; Mousavi & Eldredge, 2025); these return point estimates that regress toward the conditional mean, and classical assimilation (Raissi et al., 2020; Cai et al., 2021) recovers fields per case without amortization. Generative counterparts condition diffusion through sampling-time guidance (Shu et al., 2023; Huang et al., 2024; Amorós-Trepat et al., 2026; Chung et al., 2023; Rozet & Louppe, 2023), pretrained score models (Li et al., 2024), or amortized training (Zhuang et al., 2025; Liu & Thuerey, 2024), almost exclusively on uniform 2D grids; the 3D exceptions pay the voxel cost (Oommen et al., 2026; Lienen et al., 2024), and latent alternatives (Du et al., 2024; Zhou et al., 2025; Lino et al., 2025) inherit the autoencoder's spectral floor (Rafiq & Nair, 2026; Glaws et al., 2020).

Probabilistic evaluation of reconstruction. Although this literature routinely draws ensembles, it almost never scores them: evaluation stops at pointwise error and physics statistics, and several works claim calibrated uncertainty on the evidence of standard-deviation maps alone (Wang et al., 2025; Parikh et al., 2026; Yang et al., 2025). Proper scoring rules and calibration diagnostics (Gneiting & Raftery, 2007) appear only at the margins: a latent-diffusion boundary-layer reconstruction reports a rank histogram and finds its ensemble underdispersed (Rybchuk et al., 2023), kilometer-scale generative weather assimilation scores CRPS and spread–skill and likewise finds miscalibration (Manshausen et al., 2025), graph-diffusion urban assimilation reports CRPS on one fixed geometry (Giral et al., 2026), and sensor-time-history decoders report CRPS and coverage for 2D slices (Gao et al., 2026). To our knowledge no prior work demonstrates a calibrated distribution for 3D reconstruction from scattered point sensors under a proper scoring rule; every scored precedent in any adjacent setting is underdispersed. We therefore report fair CRPS, spread–error, coverage, and rank histograms for our model and every baseline that can produce an ensemble, with paired per-snapshot tests.

Function-space generative models and 3D scaling. Diffusion and flow matching have been lifted to function spaces with well-posedness guarantees (Lim et al., 2025; Pidstrigach et al., 2024; Kerrigan et al., 2024b; Franzese et al., 2023; Shi et al., 2024; 2025), with guided variants for inverse problems (Yao et al., 2025; Baldassari et al., 2023; Kerrigan et al., 2024a) and physics instantiations (Wang et al., 2026; Bond-Taylor & Willcocks, 2024), but demonstrations are one- and two-dimensional with FFT backbones tied to uniform grids. Deterministic surrogates separately reach millions of mesh points through graph, linear-attention, latent-grid, and anchored-attention designs (Pfaff et al., 2021; Hao et al., 2023; Wu et al., 2024; Li et al., 2023b; Luo et al., 2025; Alkin et al., 2024; 2025), establishing that locality is what scales; none of that line is generative. LOCUS keeps the function-space formulation while transferring those scaling ingredients into the conditioning pathway itself, and generates directly in ambient space on the measurement discretization with no compression stage.

3 Method

3.1 Setup: rectified flow on point sets with a Gaussian-process source

Let $u : \Omega \to \mathbb{R}^C$ be a physical field on $\Omega \subset \mathbb{R}^3$, drawn from a distribution over fields (turbulent-flow snapshots, RANS solutions over varying geometry). It is observed through a measurement operator returning readings $\mathcal{O} = \{(\boldsymbol{x}_j,\, c_j,\, y_j)\}_{j=1}^{M}$ with $y_j = u_{c_j}(\boldsymbol{x}_j) + \varepsilon_j$, where $\boldsymbol{x}_j$ is a sensor location, c_j indexes the observed quantity, and ε_j is noise. This covers locations scattered arbitrarily or confined to a surface, observed quantities forming a subset of the target channels or different quantities altogether, and known covariates entering as additional tokens. The goal is to sample an approximation to $p(u \mid \mathcal{O})$ at arbitrary query locations $\{\boldsymbol{y}_i\}_{i=1}^{N}$ that need not coincide with any training discretization; how well that approximation holds is an empirical question we answer with calibration diagnostics rather than assume.

LOCUS models the conditional distribution with a rectified flow (Liu et al., 2023; Lipman et al., 2023; Albergo & Vanden-Eijnden, 2023) defined directly on field values. The source distribution is a Gaussian process rather than pointwise white noise: a draw $u_0 \sim \pi_0$ evaluated at coordinates $\{\boldsymbol{y}_i\}$ is produced by a random-feature expansion (Rahimi & Recht, 2007),

$$u_0(\boldsymbol{y}) = \sqrt{\tfrac{2}{F}} \sum_{f=1}^{F} w_f \cos\!\big(\boldsymbol{\omega}_f^{\top} \boldsymbol{y} + b_f\big), \qquad w_f \sim \mathcal{N}(0, 1), \tag{1}$$

with frequencies $\boldsymbol{\omega}_f$ drawn once from a Gaussian whose scale sets the prior correlation length. Evaluating equation 1 at any two point sets yields marginals of one consistent function-valued source, which makes training and sampling coherent across discretizations. Given a data field u_1 at the same points, the state interpolates linearly, $u_t = (1-t)\, u_0 + t\, u_1$, and a velocity network v_θ regresses the constant displacement:

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{t,\, u_0,\, u_1,\, \mathcal{O}}\ \frac{1}{|\Omega|} \int_{\Omega} \big\| v_\theta\big(t,\, u_t(\boldsymbol{y}),\, \boldsymbol{y};\, \mathcal{O}\big) - \big(u_1(\boldsymbol{y}) - u_0(\boldsymbol{y})\big) \big\|_2^2 \, d\boldsymbol{y}. \tag{2}$$

The objective is a domain integral, and this removes the resolution barrier: equation 2 admits an unbiased Monte Carlo estimate over a random subset of query points at every step. LOCUS trains on roughly 2% of the discretization per step (3.9×10^4 of 1.95×10^6 points on the turbulence benchmark), so training memory is set by the subset size, not the resolution; voxel U-Nets, latent autoencoders, and patchified transformers do not admit this estimator, because their graphs couple all grid locations. Sampling integrates the learned ODE from a fresh source draw with a uniform Euler scheme; rectified-flow paths are near-straight, so 2–16 evaluations suffice against the hundreds used by guided diffusion (Li et al., 2024; Huang et al., 2024), and repeated integration under the same $\mathcal{O}$ yields an ensemble.

3.2 Locality-factored conditioning

The velocity network must expose the observations to every query point without incurring an all-to-all term in N or a dense volumetric representation. LOCUS factors the conditioning pathway by locality (Figure 1).

Each observation $(\boldsymbol{x}_j, c_j, y_j)$ is embedded as a token from a Fourier positional encoding of $\boldsymbol{x}_j$ (Tancik et al., 2020), the reading y_j, and a learned embedding of the quantity index c_j. A small array of S learnable latents cross-attends to the M sensor tokens (Jaegle et al., 2021) at cost $\mathcal{O}(SM)$, refined by self-attention blocks that re-attend to the sensor tokens

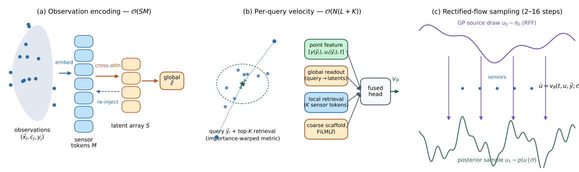

Figure 1: The LOCUS architecture. Sensor tokens are compressed by cross-attention into a small latent array (global pathway, $\mathcal{O}(SM)$), which re-injects context back into the sensor tokens. Each query assembles its velocity from its point features, a readout from the latent array, and an importance-weighted retrieval over its K nearest sensor tokens (local pathway, $\mathcal{O}(NK)$). No dense volumetric representation is formed; cost is linear in the number of queries, and training estimates the loss on a random ∼2% point subset per step.

between rounds; the mean-pooled array provides a global summary, and a final pass writes that context back into the sensor tokens.

Each query $\boldsymbol{y}_i$ then fuses three components: a point feature encoding its position, the current state $u_t(\boldsymbol{y}_i)$, and t; a global readout cross-attending from a coordinate-conditioned decoder token to the latent array at cost $\mathcal{O}(L)$; and a local retrieval over the K nearest sensor tokens under an importance-warped metric $d(\boldsymbol{y}_i, \boldsymbol{x}_j)^2 = \|\boldsymbol{y}_i - \boldsymbol{x}_j\|_2^2 - \beta_j$, combined with normalized RBF weights. The learned per-sensor bias β_j lets informative sensors extend their influence beyond their Euclidean neighborhood, which matters when sensing is anisotropic (surface-only layouts) or partially corrupted. A FiLM-modulated coarse prediction from the global summary enters as a gated residual, and a small MLP head produces the velocity. Retrieval and the k-nearest-neighbor search run as streaming kernel reductions (Charlier et al., 2021), so the $[N \times M]$ distance matrix is never materialized. Total cost per evaluation is $\mathcal{O}(SM + NL + NK)$ with no term coupling query points to each other: compute is linear in N, and the same trained network evaluates on 10^4 or 2×10^6 points unchanged.

3.3 Training objective and sampling cost

Two additions to equation 2 matter in practice. First, t is drawn logit-normally, concentrating capacity at intermediate noise levels (Ma et al., 2024). Second, generative models of turbulence exhibit spectral bias, reproducing large scales and attenuating the inertial range (Rahaman et al., 2019); following the observation that binned spectral penalties succeed where naive FFT losses do not (Chakraborty et al., 2026), each batch appends a 32^3 lattice block to the query set, and from the endpoint estimate $\hat{u}_1 = u_t + (1-t)\, v_\theta$ we penalize the squared difference of shell-averaged log power against u_1 over 12 radial bins, weighted by t. The total loss is $\mathcal{L}_{\mathrm{FM}} + \lambda_{\mathrm{sp}}\mathcal{L}_{\mathrm{sp}}$ with $\lambda_{\mathrm{sp}} = 0.02$, and Section 4.1 reports the resulting trade-off with λ_{sp} as the control.

At sampling time every quantity depending only on the observations and query coordinates—the sensor branch, the neighbor geometry, and the coordinate-conditioned readout—is computed at the first ODE step and reused exactly thereafter, so the marginal cost of a step falls to the fusion head: a full 125^3 sample costs 4.5 s at 4 evaluations and 5.1 s at 16, making high-step operating points nearly free.

Amortized measurement operators. Conditioning is amortized by randomizing the measurement operator during training: each step draws the sensor count and layout afresh, and with configurable probability readings are corrupted with Gaussian noise of random amplitude, contiguous sensor regions are removed, and entire observed channels are dropped. The network thus learns one conditional velocity field over a family of operators, and at test time a single model serves that family without retraining or guidance; Section 4.3 additionally probes operators outside the training ranges. Known covariates (Mach, incidence) enter as parameter tokens exempt from corruption. Observed values can be enforced on the generated field at sampling time; we enforce for noiseless protocols and disable under sensor noise, where interpolating corrupted readings is undesirable. Training details are in Appendix B.

4 Experiments

Evaluation protocol. Turbulence snapshots are strongly autocorrelated in time, and splitting consecutive DNS frames at random—standard practice—makes every validation frame a near-duplicate of a training frame (frame-to-frame correlation 1.00 at unit separation). Under such a split all methods, ours included, achieve errors two to three times lower, and the ranking measures interpolation rather than reconstruction. All turbulence results therefore use spatial holdout: training on three 125^3 regions of the DNS, evaluation on a fourth, disjoint region, with temporally blocked splits inside the training set. Probabilistic metrics use K=8 ensembles on matched snapshots with identical seeded sensor draws for every method, compared by paired per-snapshot confidence intervals. Exact octahedral symmetry augmentation (48 signed axis permutations, discretization-exact on cubic grids) is supplied to every method. [TODO: Temporally blocked same-region companion evaluation (retrains queued).]

Datasets. Three domains, each exposing one failure mode of existing methods (preprocessing in Appendix C). (i) Forced isotropic turbulence (Li et al., 2008): four disjoint 125^3 cutouts of the 1024^3 DNS (1.95×10^6 points, velocity and pressure, 50 strided snapshots each), three for training and the fourth for evaluation. (ii) SHIFT-WING (Luminary Cloud, 2026): adaptively meshed RANS over transport-aircraft wing–fuselage variants at Mach 0.5–0.85; 673 cases (600/73), each 4×10^5 volume and 6.5×10^4 surface nodes carrying pressure and wall shear, geometry varying across cases and entering only through the observation point cloud. (iii) FireBench (Wang et al., 2024): wildfire LES at two wind speeds; we extract the fire-front region (3.7×10^6 points; velocity, potential temperature, fuel density; 120 snapshots).

Baselines. We compare against the routes identified in Section 1, configured for fairness rather than convenience (Appendix D). Latent route: a 3D convolutional autoencoder with rectified flow in its latent space (LDM-style; Rombach et al., 2022), sensor-conditioned through a point encoder, parameter-matched to LOCUS, trained in two stages on identical data, splits, and augmentation; it stands in for the latent-diffusion family (Du et al., 2024). Voxel route: Gen4Turb (Oommen et al., 2026), a published diffusion U-Net with released code, run at its native 120^3 on the same data, wall-clock-matched, and additionally retrained under our exact observation protocol. Deterministic point route: Senseiver (Santos et al., 2023), trained with identical randomization and augmentation. [TODO: Point-generative baselines (SiT-point, CoNFiLD-class) on the turbulence benchmark and wing; gappy-POD anchor.]

Metrics. We report relative L_2 of the ensemble mean and of single samples (the latter measures whether individual realizations are physical rather than over-smoothed), band-resolved spectral ratios, fair CRPS (Gneiting & Raftery, 2007), spread–error ratio, and central 90% coverage, in per-field standardized units (Appendix E). Calibration estimates from K=8 ensembles are coarse; a K=32 check shows they are conservative for every method (ours $0.55 \to 0.66$ coverage, latent baseline $0.45 \to 0.52$) with the ranking unchanged.

4.1 Accuracy and calibration on spatially disjoint turbulence

Table 1 reports the head-to-head, with paired per-snapshot statistics over all 50 test snapshots. On the proper score LOCUS wins or ties: without the spectral penalty its CRPS beats the latent baseline significantly (-0.014, 95% CI $[-0.026, -0.002]$), and the primary λ_{sp}=0.02 model is indistinguishable (-0.011, CI $[-0.025, +0.004]$), with TOST equivalence inside a ± 0.03 margin (p=0.004). On ensemble-mean relative L_2 the latent baseline holds a small but significant edge ($+0.029$, CI $[+0.009, +0.050]$), bounded below 0.05 by TOST (p=0.02). Calibration separates decisively: spread–error and coverage differences of $+0.097$ $[+0.075, +0.118]$ and $+0.087$ $[+0.069, +0.106]$ ($t \approx 9$; for λ_{sp}=0, $+0.29$ and $+0.20$ at $t \approx 28$). [TODO: Add training-seed replicates.] No model here is calibrated in the absolute sense—coverage stays below nominal, as in every scored precedent, which is why we call these predictive rather than posterior distributions—but LOCUS is closest by a wide margin (0.82

Table 1: Isotropic turbulence, spatially disjoint evaluation region (1.95M points, two velocity channels observed at 1% of points, K=8, all 50 held-out snapshots). Spread–error and coverage ideals are 1.0 and 0.90. Senseiver is deterministic (CRPS reduces to MAE, no spread). Gen4Turb is retrained under our observation protocol at its native 120^3; under its native shared-mask protocol, which observes all four channels (twice the information), it reaches 0.354/0.204 on an 8-snapshot subset. [TODO: SiT-point and CoNFiLD rows: training/pipeline running.]

Method	Rel. L_2 mean	Rel. L_2 sample	CRPS	Spread/Err.	Cov.$_{90}$
Senseiver (deterministic)	0.722	—	0.542	—	—
Gen4Turb (voxel diffusion)	0.713	0.735	0.441	0.28	0.25
Latent flow matching	0.564	0.629	0.302	0.53	0.45
LOCUS (λ_{sp}=0, NFE 4)	0.623	0.811	0.288	0.82	0.65
LOCUS (NFE 2)	0.582	0.641	0.312	0.37	0.33
LOCUS (NFE 4)	0.593	0.711	0.291	0.63	0.54

spread–error, 0.65 coverage at λ_{sp}=0, against 0.53 and 0.45). The voxel-diffusion baseline does not close the gap even retrained under the matched protocol: 0.713 relative L_2 with a severely underdispersed ensemble (coverage 0.25), and retraining moved accuracy by less than 0.01—a mask-channel voxel U-Net extracts little of the information in scattered point observations. Its stronger shared-mask number (0.354) requires observing all four channels, which our protocol deliberately does not grant.

Two knobs expose a distortion–dispersion frontier. Step count: two evaluations give the sharpest single samples (0.641, best of any generative method), four are the CRPS optimum, and dispersion rises monotonically with further steps (spread–error 0.92 at NFE 16 for λ_{sp}=0, at flat CRPS; Appendix F), which the per-solve conditioning cache makes nearly free. Spectral weight: raising λ_{sp} from 0 to 0.02 improves single-sample error and inertial-band fidelity at a measured cost in spread, and at 0.05 the accuracy gain saturates while dispersion degrades further. We report λ_{sp}=0.02 as the primary model; no single point on the frontier is best for every use.

Two confounds are closed by ablation (Appendix F): doubling the per-step query budget from 2% to 4% does not improve held-out loss (three seeds), so subsampled supervision is not the binding constraint even though the latent baseline consumes roughly 50× more supervised values per step; and octahedral augmentation is the operative one, with continuous rotations, translations, and proper-rotation restrictions all inside run-to-run variance.

What the uncertainty tracks. Figure 2 shows the reconstruction and its spread directly; Appendix F adds pointwise diagnostics. The predictive standard deviation separates the channels the sensors constrain from those they do not: on the observed U_x it averages 0.09 against RMS error 0.17, while on the unobserved U_y it rises to 0.87 against RMS error 0.95. Two negative results are worth stating. The spread–error relation is monotone but above the diagonal, the pointwise form of the underdispersion the aggregate metrics report; and the standard deviation is nearly independent of distance to the nearest sensor—flat at 1% coverage and still flat at 0.1%, where that distance reaches eight cells. We expected a profile growing away from the data and do not find one: spread here is governed by which channels are observed and by local flow structure, not by distance to the nearest reading, consistent with a conditioning design in which every query also reads a global summary of the sensor set.

Small-scale fidelity. Energy spectra, structure functions and velocity-gradient PDFs of single samples (Figure 4, Appendix F) test the premise behind avoiding an autoencoder. One estimator detail is load-bearing, so we state it first: the cutouts are sub-blocks of a larger DNS and are not periodic, so an unwindowed FFT spreads the edge discontinuity into a broadband floor. On our data the raw spectrum of the truth falls only 2.4× between k=32 and the Nyquist wavenumber, and a deliberately smooth Gaussian-process draw—whose energy there is zero to machine precision—registers the same floor. A variance-preserving

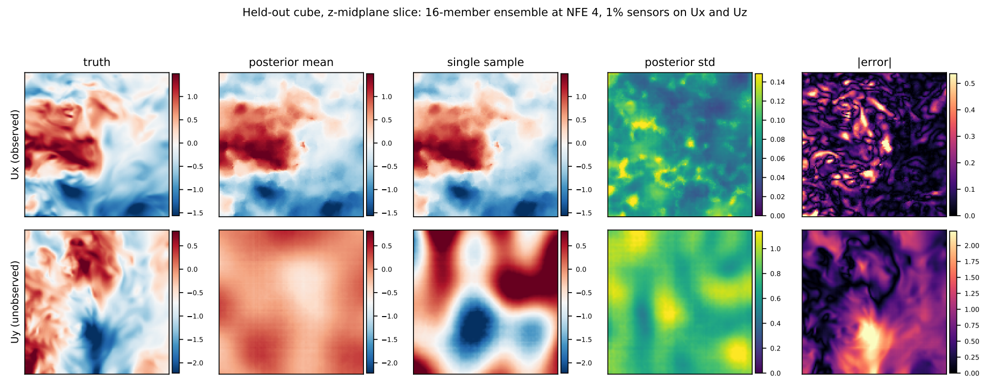

Figure 2: Reconstruction and uncertainty on the held-out cube (z-midplane slice, 16-member ensemble at NFE 4, 1% sensors on U_x and U_z). Top: an observed channel, recovered sharply, standard deviation an order of magnitude below the field scale. Bottom: U_y, which no sensor constrains—the ensemble mean collapses toward the prior mean while individual samples stay sharp and the standard deviation rises to the field scale, so the model reports that it does not know. The |error| column tracks the standard-deviation column throughout (pointwise correlation 0.60).

Hann taper restores a 5.8×10^5 decay and reduces that same draw to 5×10^{-5} of the DNS inertial-range energy. Ratios computed without the taper are ratios of two floors and tend to unity for every method; all spectral numbers we report use the windowed estimator.

Measured this way the comparison splits by band and neither model is uniformly better. The latent baseline holds more inertial-range energy than LOCUS (0.62 vs. 0.41 for U_x over $k \in [8, 31]$) but collapses in the dissipation range, retaining 0.07–0.09 of the DNS energy over $k \in [32, 45]$ against our 0.40–0.46: the autoencoder spectral floor (Rafiq & Nair, 2026; Glaws et al., 2020) appearing where it was predicted. Our own high-wavenumber content is not recovered turbulence either—the spectrum flattens into a plateau instead of continuing to decay, the signature of broadband noise, reaching 3.5–4.8× the DNS energy on pressure. Ambient generation therefore avoids the compression floor at the smallest scales while underproducing the inertial range, latent compression does the opposite, and neither reproduces the DNS spectrum from 1% sensors; Appendix F localizes our deficit to the channels the sensors do not constrain.

4.2 The cost of resolution

Figure 3 reports measured peak memory and wall-clock for training steps and full-field sampling from 64^3 to 1024^3; every out-of-memory marker is an observed failure on an 80 GB H100. The axis separates two computational families. Grid methods scale cubically: the convolutional autoencoder anchoring the latent baseline trains at 50.9 GB for one 512^3 field and fails at 640^3 at batch size 1 (at production batch size the wall falls between 256^3 and 320^3), and the voxel diffusion model fails at 320^3 at batch size 1 while already consuming 57.5 GB at 120^3. Point methods are flat: LOCUS trains at 39.9 GB and 0.44 s per step and samples at 1.4 GB peak at every resolution tested, and Senseiver shares the profile (19.4 GB train, 1.2 GB inference)—flatness is a property of point computation, not of our model alone. Within that family LOCUS is the only generative member, and the deterministic one pays a factor of 1.3 in accuracy and cannot produce an ensemble.

The annotations give the mechanism: a voxel or convolutional model is structurally committed to supervising every grid cell at every step—its memory is the supervision—whereas LOCUS chooses its budget independently of resolution, and the ablations above show that budget saturates. The honest converse is that grid computation is cheaper below the crossover ($\sim$$240^3$ for inference): a convolutional decode of a 125^3 field costs 6 ms where our chunked point evaluation costs 4.5 s. Stated precisely, the claim is constant memory with respect to the output discretization and linear-time inference, not constant total cost; the regime

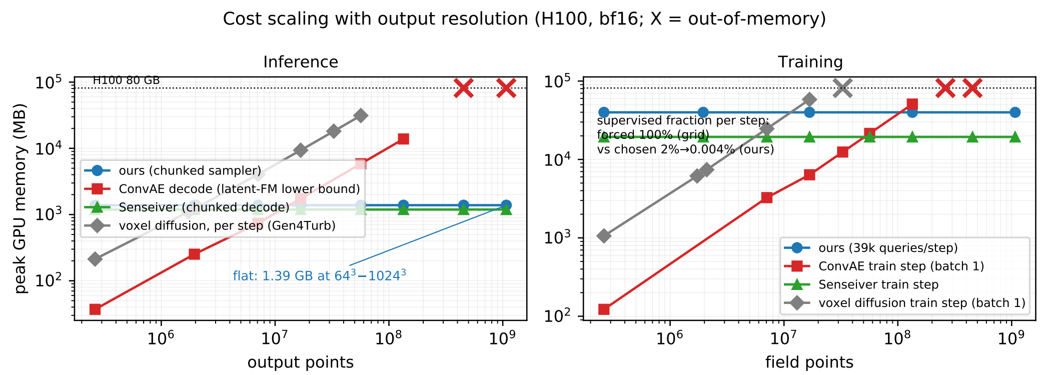

Figure 3: Measured peak memory versus output resolution (H100, bf16, log–log; OOM markers are measured failures, not extrapolations), for full-field sampling (left) and training steps (right). Grid methods (voxel diffusion, convolutional AE) scale cubically and fail at 320^3 and 640^3; point methods (LOCUS, Senseiver) are flat to 1024^3 (10^9 points). The annotation gives the fraction of the field supervised per training step: 100% forced for grid methods, a fixed 39k-point budget for LOCUS (2% at 125^3, 0.004% at 1024^3). Wall-clock panels are in Figure 5.

Table 2: Wildfire LES (3.7M points, 5 fields) under realistic measurement operators (relative L_2 of ensemble mean / fair CRPS; K=8, 8 snapshots, identical seeded operator realizations for all methods). Sensors observe the three wind components at ~1% of points. LOCUS is a single amortized model across all rows.

	LOCUS		Latent FM		Senseiver	
Operator	Rel. L_2	CRPS	Rel. L_2	CRPS	Rel. L_2	CRPS
Clean	0.456	0.234	0.724	0.356	0.779	0.443
Noise σ=0.1	0.458	0.237	0.722	0.355	0.779	0.444
Noise σ=0.3	0.461	0.248	0.724	0.356	0.779	0.445
Slab occlusion (25%)	0.511	0.275	0.720	0.355	0.780	0.443
Channel dropout (u only)	0.603	0.355	0.719	0.309	0.908	0.514

where LOCUS is the only option is the one the applications occupy—resolutions above ~300^3, unstructured discretizations, and sensing geometries no grid can represent. The constancy of the training graph also raises the right follow-up question, whether the model trained at one discretization transfers as a function-valued sampler to finer and irregular ones; a direct discretization-transfer experiment (training at 125^3, reconstructing a native 500^3 cutout and irregular subsets of it) is in preparation as a core test of the function-space claim. [TODO: Discretization-transfer experiment: 125^3-trained model evaluated on a 500^3 JHTDB cutout and on irregular/coarsened point sets, with convergence of spectra and error across discretizations.]

4.3 Deployed sensing: corrupted operators and native geometry

Table 2 evaluates the regime the introduction argued for, on wildfire LES at 3.7M points. One amortized LOCUS wins every cell, with qualitative margins: 0.3σ sensor noise costs one percent relative, removing a 25% slab of sensors twelve percent, and dropping two of three wind components a third. Its worst case (0.603) still beats either baseline's best case under clean, complete observations (0.719 and 0.779). One row is genuinely out of distribution—training used noise amplitudes up to 0.1σ, so the 0.3σ row is triple anything seen—while the occlusion and dropout rows lie inside the amortized family, demonstrating robustness within a deliberately broad operator distribution rather than extrapolation. [TODO: Add further OOD operators: ball occlusion, sensor density outside the training range, structured arrays.]

The baselines' apparent stability across rows deserves the correct reading: their numbers barely move because they are bias-dominated, and a model that extracts little of the information in the sensors loses little when that information is corrupted. The latent baseline reconstructs at 0.72 relative L_2 under every operator (predicting the zero field scores 1.0), and its CRPS improves under channel dropout only because its spread is large where its mean is wrong. LOCUS's errors respond to the information content of the observations because the model uses that information; graceful, monotone degradation, not invariance, is the behavior a practitioner should want. The latent baseline's dispersion ratio here (0.87) exceeds ours (0.76), but at more than twice the CRPS; the proper score, integrating both, favors LOCUS by 34% in every condition. A clean-trained control isolates what operator amortization actually buys: trained without any corruption, the control matches the robust model under noise and slab occlusion (0.458 vs. 0.461 at 0.3σ; 0.513 vs. 0.511 under occlusion)—robustness to those perturbations comes from the conditioning architecture, not the training distribution—but collapses under channel dropout (0.743 vs. 0.603), the one operator that changes the structure of the observation set. Amortized operator training is what covers that structural shift, and it costs nothing in the clean condition (0.455 vs. 0.456).

Cross-variable inference falls out of the same interface: observing only wind, the model recovers unobserved fuel density at 0.044 relative L_2 and potential temperature at 0.49, degrading only to 0.064 and 0.52 when two of the three wind components are also removed (Appendix F). The randomized observed-channel embedding is what makes a wind-only, single-component operator an interpolation rather than an extrapolation for the conditioning network. The uncertainty is placed where the physics is ambiguous rather than uniformly: averaged over the hottest 5% of the domain the predictive standard deviation is 7–25× its value in the ambient half (for potential temperature, 1.32 against 0.05), and Figure 7 shows it tracing the fire front in the fuel bed, which the model never observes.

Surface-only observation on native geometry. The wing exercises two capabilities no grid-based generative method has: an unstructured mesh whose geometry changes across cases, and observations confined to the body surface. From 640 surface sensors (512 pressure taps, 128 shear gauges, with Mach and incidence as parameter tokens), LOCUS recovers the volumetric pressure-coefficient field on 8 held-out geometries at 0.126 relative L_2 and velocities at 0.53–0.69 (CRPS 0.18, spread–error 0.68, coverage 0.38; K=4, NFE 4), with predictive standard deviation concentrating in the boundary layer and wake—the regions genuinely undetermined by surface data. The importance-warped retrieval is what lets surface sensors inform off-body queries, since under a plain Euclidean metric most volume points have no nearby sensors; we treat its necessity as a hypothesis for direct ablation. Grid-based baselines are structurally excluded here, which is the point. [TODO: Wing rows (Senseiver-pool, SiT-point, gappy POD), Euclidean-retrieval ablation, tap sweep, slice gallery.]

5 Conclusion

Generative reconstruction of 3D fields has been blocked not by modeling capacity but by a structural mismatch: generative architectures compute globally on dense discretizations, while the problem is defined on scattered points and needs only local and summary information at each. LOCUS resolves it by carrying rectified flow directly on field values with a Gaussian-process source, locality-factored conditioning, and subsampled function-space training. The measured consequences: costs flat from 64^3 to 1024^3 where voxel and autoencoder pipelines fail at 320^3–640^3; accuracy competitive with a strong latent baseline on spatially disjoint turbulence, with the best-calibrated predictive distributions of any method evaluated; uniform dominance under corrupted sensing at 3.7M points; and volumetric predictions from surface-only sensors on varying unstructured geometries.

Limitations are equally measurable. Coverage stays below nominal even at our best-calibrated operating point (0.65 against 0.90)—best-calibrated among evaluated methods, not calibrated; single-snapshot reconstruction ignores temporal information that assimilation exploits; per-domain training is still required; and constant-memory inference pays a linear time cost, making LOCUS the slowest method in absolute terms at large resolution (41 minutes for a 10^9-point sample). Sharpening calibration, conditioning on sensor time histories, and pretraining across flow families are the natural continuations.

Reproducibility statement

All experiments use publicly available data: the JHTDB isotropic turbulence database, the SHIFT-WING dataset (CC-BY-NC-4.0), and the public FireBench store; exact cutout indices, crop definitions, subsampling seeds, and normalization are specified in Appendix C. Architecture and training hyperparameters for LOCUS and all baselines are listed in Appendices A–D together with the configuration files used for every run. Code, configurations, and dataset preparation scripts will be released. [TODO: Anonymous code link at submission.]

AI use statement

[TODO: Complete per ICLR 2027 policy before submission: disclose the use of AI assistance for literature search, code development, and manuscript drafting, with author verification of all content.]

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A Architecture details

The velocity network used for all experiments has 6.51M trainable parameters (parameter-matched to the latent flow-matching baseline's two stages combined). Sensor tokens and queries use Fourier positional encodings with 32 bands and maximum frequency 64. The global pathway compresses M sensor tokens into $S{=}128$ latents of width 256 with 8-head cross-attention, refined by 4 latent self-attention blocks with feed-forward multiplier 4; sensor re-injection cross-attention runs after every block, and a final back-attention pass writes latent context into the sensor tokens. The per-query readout attends from a coordinate-conditioned decoder token to the latent array. The local pathway gathers the $K{=}16$ nearest sensor tokens under the importance-warped metric of Section 3.2 with a learnable RBF bandwidth; a training-time ablation on a trained checkpoint found K=64, 32, and 16 indistinguishable in reconstruction error (relative L_2 0.273/0.273/0.272), so the smallest is used. Neighbor search and RBF aggregation run as streaming KeOps reductions (Charlier et al., 2021) with query chunking (2,048 queries per chunk for the global readout, 4,096 for decoding). The source GP uses 256 random Fourier features with correlation length 0.15 in normalized coordinates.

Two implementation details matter at our scale. First, cross-attention over tens of thousands of variably valid sensor tokens with a key-padding mask forces the memory-efficient (CUTLASS) attention backward, which at $M \approx 39$k keys is $\sim 70\times$ slower at the kernel level than the flash backward; selecting each item's valid keys by boolean mask and running unmasked attention per item is mathematically identical (max deviation 4×10^{-6}) and doubles training throughput ($0.88 \rightarrow 0.41$ s per step, compiled). Second, during ODE sampling, all sensor-branch activations, neighbor geometry, and the coordinate-conditioned readout are cached after the first step (Section 3.3); the cache is exact (bitwise-zero output difference) and reduces a 16-step 125^3 solve from 69.5 s to 5.1 s.

B Training details

All models train on single 80 GB H100 GPUs in bf16 autocast with an exponential moving average of weights (decay 0.9995) used for evaluation, Adam at learning rate 10^{-4}, weight decay 10^{-6}, and batch size 20 point-cloud samples. Turbulence: 6,000 epochs ($\sim$20 h wall-clock; the latent baseline's two stages total $\sim$21 h, so total budgets are matched), 39,062 query points per sample per step, per-channel sensor counts drawn log-uniformly between 0.1% and 1% of points, observed channels $\{U_x, U_z\}$, t logit-normal, λ_{sp}=0.02 with a 32^3 spectral block and 12 bins. Wildfire: 3,500 epochs, observed channels $\{u, v, w\}$, sensor counts 0.03–1%, operator amortization with noise amplitude up to 0.1σ, slab occlusion probability 0.2, and observed-channel dropout probability 0.3; lateral-reflection augmentation. Wing: 4,000 epochs, surface pool of 512 pressure + 128 shear sensors drawn per step from the 6.5×10^4-node skin, Mach and incidence tokens, spanwise-reflection augmentation. The forward pass is compiled; compilation preserves checkpoint compatibility.

C Datasets and preprocessing

Isotropic turbulence. Four 125^3 cutouts at disjoint spatial locations of the JHTDB isotropic1024coarse DNS, 50 snapshots each at a stride of 100 database steps, fields (U_x, U_y, U_z, p), per-field standardized. Training uses cutouts 1–3 (150 snapshots); all reported evaluation uses cutout 4. Within the training cutouts, temporal blocks are kept contiguous. The leakage analysis motivating this protocol (Section 4) was run on 617 consecutive frames of a single cutout: frame-to-frame correlation is 1.00 at unit separation and 0.67 at separation 100, and a shuffled split placed every validation frame within one frame of a training frame.

SHIFT-WING. 673 converged RANS cases spanning the dataset's Mach blocks (0.5–0.85), incidence 0.02–3.99°, and 40 geometry parameters (aspect ratio 7.5–11, sweep 25–37°). Each case is reduced to 4×10^5 near-body volume nodes (subsampled from the adaptive mesh) and the full 6.5×10^4-node surface pool carrying C_p and the three wall-shear components as distinct observed-quantity indices; Mach and incidence enter as parameter tokens with dedicated indices. Split 600/73 by case; physical-coordinate spanwise reflection augments training.

FireBench. From the public FireBench store, the two LES runs at 10 and 12 m/s driving wind. The 3D extraction crops a $152 \times 126 \times 192$ region around the fire front (streamwise 250–706 m at 3 m, full lateral extent at 2 m, 0–192 m altitude at 1 m), avoiding the empty sky aloft, with fields $(u, v, w, \theta, \rho_f)$; 60 snapshots per run at stride 2, merged to 120, temporally blocked split with a 10-frame guard gap.

D Baseline configurations

Latent flow matching. A 3D convolutional autoencoder (3 levels, $\sim 16^3 \times 48$ latent) trained for 5,000 epochs on full fields, followed by 10,000 epochs of rectified flow in the latent space with a point-encoder conditioning branch over the identical sensor draws; combined parameter count 6.53M, matched to ours. Identical data, splits, augmentation, and sensor protocol. The two-stage structure gives this baseline roughly 50× more supervised field values per step than LOCUS; the supervision ablation in Appendix F addresses the resulting confound.

Gen4Turb. The released code and architecture of Oommen et al. (2026) (elucidated-diffusion U-Net, channel dimension 16, self-conditioning, mask-channel conditioning), trained on our turbulence data at its native 120^3 crop of the 125^3 region, with σ_{data} taken from its own preprocessing. Its training script defaults to a compute budget an order of magnitude beyond every other method; we checkpoint every 10 epochs and report the wall-clock-matched checkpoint (epoch 4,930, matching our 20 h). Two observation protocols are evaluated: its native shared-mask protocol (all four channels observed at the same 1% locations; more information than any other method receives) and our strict protocol (two channels at independent locations). Because the strict protocol is out of distribution for the shared-mask training, we additionally retrained it under our exact observation model (independent per-channel masks, coverage drawn 0.1–1%); the retrain moved strict-protocol accuracy by less than 0.01 (0.700→0.706), and its validation-selected checkpoint is substantially worse than the budget-matched one (0.921 vs 0.706), so the table reports the budget-matched retrained model, the choice most generous to the baseline.

Senseiver. The attention-based deterministic reconstructor of Santos et al. (2023) under identical sensor randomization, augmentation, and splits, trained to its own convergence. As a deterministic model its CRPS equals its MAE, and no dispersion metrics apply.

Operator evaluations. All test-time operators (noise, slab occlusion, channel dropout) are applied to the assembled sensor sets by shared seeded code, so every method sees bit-identical corrupted observations per snapshot.

E Metric definitions

For an ensemble $\{s_k\}_{k=1}^K$ and truth y at a point, the fair CRPS estimator is CRPS $= \frac{1}{K}\sum_k |s_k - y| - \frac{1}{2K(K-1)}\sum_{k\neq l}|s_k - s_l|$, averaged over points and fields in standardized units. The spread–error ratio is the ratio of the ensemble standard deviation (RMS over points) to the RMS error of the ensemble mean; 1.0 indicates dispersion consistent with error. Coverage$_{90}$ is the fraction of points whose truth falls inside the central 90% ensemble interval. Rank histograms accumulate the rank of the truth within the ordered ensemble per point. Relative L_2 is $\|\hat{u} - u\|_2/\|u\|_2$ per field, reported for the ensemble mean and for individual samples.

F Additional results

Table 3: Spectral-weight ablation on the turbulence benchmark (matched protocol, K=8, 8 snapshots). Ratios are single-sample inertial-band spectral power relative to truth at NFE 2. The weight trades single-sample sharpness against posterior dispersion.

λ_{sp}	Rel. L_2 mean	Rel. L_2 sample (NFE 2)	CRPS	Spread/Err.	Cov.$_{90}$	Inertial ratio
0	0.612	0.730	0.320	0.79	0.65	0.43
0.02	0.592	0.651	0.333	0.64	0.55	0.48
0.05	0.619	0.676	0.340	0.68	0.59	0.48

Table 4: Sampling-step (NFE) sweep for the λ_{sp}=0 model (K=4): dispersion improves monotonically with step count at flat CRPS; the conditioning cache makes the sweep nearly cost-free (4.4–5.1 s per 125^3 sample).

NFE	Rel. L_2 (mean)	CRPS	Spread/Error	Coverage$_{90}$
2	0.607	0.299	0.65	0.41
4	0.635	0.294	0.82	0.51
8	0.653	0.294	0.89	0.54
16	0.662	0.295	0.92	0.55

Table 5: Wildfire per-field relative L_2 (ensemble mean) for LOCUS: cross-variable inference of unobserved thermochemical fields from wind observations.

Operator	u	v	w	θ (unobs.)	ρ_f (unobs.)
Clean (wind observed)	0.442	0.726	0.583	0.487	0.044
Channel dropout (u only)	0.470	1.026	0.932	0.524	0.064

Table 6: Robust-vs-clean training ablation on the wildfire operator matrix (relative L_2 / CRPS, K=8, 8 snapshots). The clean-trained control saw no corrupted sensors during training.

Operator	LOCUS (operator-amortized)	LOCUS (clean-trained)
Clean	0.456 / 0.234	0.455 / 0.231
Noise σ=0.1	0.458 / 0.237	0.456 / 0.234
Noise σ=0.3	0.461 / 0.248	0.458 / 0.246
Slab occlusion (25%)	0.511 / 0.275	0.513 / 0.277
Channel dropout (u only)	0.603 / 0.355	0.743 / 0.436

Per-channel spectra and the unconstrained-channel deficit. Table 7 localizes the inertial-range gap. On observed channels LOCUS retains about 0.40 of the DNS inertial energy; on U_y, which no sensor constrains, it retains 0.03–0.06, against 0.67 for the latent baseline, whose convolutional decoder emits textured fields regardless of what the conditioning specifies. Raising the solver-step count from 2 to 64 moves U_y only from 0.026 to 0.057, so the sampling budget is not the binding constraint. The prior row identifies what is: our random-feature source holds 5×10^{-5} of the DNS inertial energy, so a channel the observations leave free stays close to an essentially empty draw. Two targeted changes are under test—a source whose radial frequency density gives $E(k) \sim k^{-5/3}$ rather than a single smooth scale, and a windowed spectral objective, the untapered version of which reads its top shells $\sim$100× too high and is correspondingly weak. [TODO: Report both retrains (prior, windowed loss) here.]

Augmentation and supervision ablations (turbulence). With octahedral augmentation as the base (best held-out flow-matching loss 0.7475), adding translations and restriction to proper rotations (0.7515), continuous rotations (0.7783), or narrowing the latent array (0.7648) all fall within within-run fluctuation (0.05–0.11); no intervention improved on plain octahedral. Without any augmentation, all methods—including both baselines—diverge on the spatially disjoint region, confirming that at $\sim$150 training snapshots symmetry augmentation is what makes cross-region generalization possible at all. Doubling the query budget to 78k points per step (three seeds: 0.756, 0.790, 0.767) does not improve on 39k (0.7475). A heavy-regularization arm (weight decay 10^{-2}, dropout 0.1) is likewise null (0.769–0.786), indicating the binding constraint is snapshot count, not overfitting control or per-step supervision.

Cost at the benchmark resolution. At 125^3: LOCUS trains at 0.44 s/step, 39.9 GB, and samples a full field in 4.5 s (NFE 4) at 1.4 GB; the latent baseline trains its stages at

Table 7: Per-channel spectral content of single samples (Hann-windowed, 4 snapshots): inertial $k \in [8, 31]$ / dissipation $k \in [32, 45]$ energy relative to the DNS. U_x and U_z are observed; U_y and p are not. The source prior row shows the diagnosis: it is empty at these wavenumbers, so an unconstrained channel that stays near the prior inherits that emptiness.

	U_x (obs.)	U_y (unobs.)	U_z (obs.)	p (unobs.)
LOCUS, NFE 2	0.41 / 0.48	0.026 / 0.51	0.40 / 0.42	0.26 / 4.77
LOCUS, NFE 4	0.41 / 0.45	0.025 / 0.47	0.39 / 0.40	0.26 / 4.01
LOCUS, NFE 16	0.41 / 0.45	0.046 / 0.75	0.39 / 0.40	0.27 / 3.60
LOCUS, NFE 64	0.42 / 0.46	0.057 / 0.89	0.39 / 0.40	0.27 / 3.53
Latent flow matching	0.62 / 0.09	0.67 / 0.09	0.53 / 0.07	0.21 / 0.04
Source prior (ours)	0.00005 / $< 10^{-5}$	—	—	—

8.2/3.4 GB and samples in 0.7 s at 0.25 GB; Senseiver trains at 0.17 s/step, 19.4 GB, and decodes in 0.13 s at 1.2 GB; Gen4Turb trains at 57.5 GB (production configuration, 120^3) and samples in $\sim$1 s (32 denoising steps). Below the $\sim 240^3$ crossover, grid computation is cheaper; the separation of Figure 3 is about what happens above it.

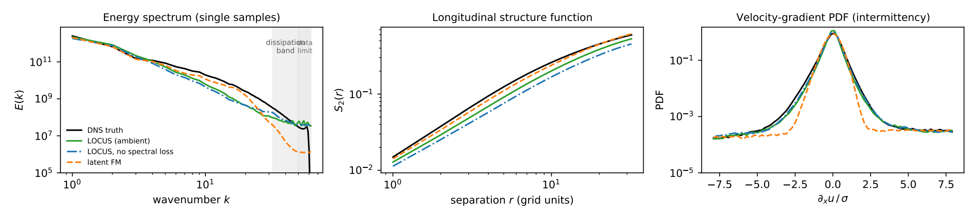

Figure 4: Turbulence statistics of single posterior samples on the held-out region (4 snapshots, NFE 4, Hann-windowed; ensemble means are deliberately not shown—averaging destroys small scales by construction). Left: shell-averaged energy spectra. The latent baseline tracks the DNS through the inertial range and then falls away sharply in the dissipation band (shaded), the autoencoder floor; LOCUS sits below the DNS through the inertial range and instead flattens into a plateau, which is broadband noise rather than recovered structure. Beyond $k \approx 50$ the DNS itself is resolution-limited (light shading). Middle: longitudinal structure functions. Right: velocity-gradient PDFs, where the latent baseline distorts the shoulders while both LOCUS variants track the DNS shape.

Pending additions. [TODO: Ordered by review severity: (i) expanded statistics — all 50 held-out snapshots, K=32 sensitivity, TOST equivalence bounds, training-seed replicates (evals running, retrains queued); (ii) point-generative baseline on the main turbulence benchmark; (iii) discretization-transfer experiment (500^3 cutout + irregular/coarsened point sets) as a core function-space test; (iv) architectural ablations: GP source vs. i.i.d. noise, global-only vs. local-only vs. both pathways, Euclidean vs. importance-warped retrieval (wing), supervision-matched training, latent-vs-ambient with shared conditioning; (v) spectra figure: add error bars over snapshots and a seed-replicate band; (vi) temporally blocked same-region evaluation; (vii) further OOD operators on wildfire (ball occlusion, out-of-range sensor density, structured arrays); (viii) compact-query backbone trial (running); (ix) wing baselines, tap-count sweep, qualitative galleries, rank-histogram figure.]

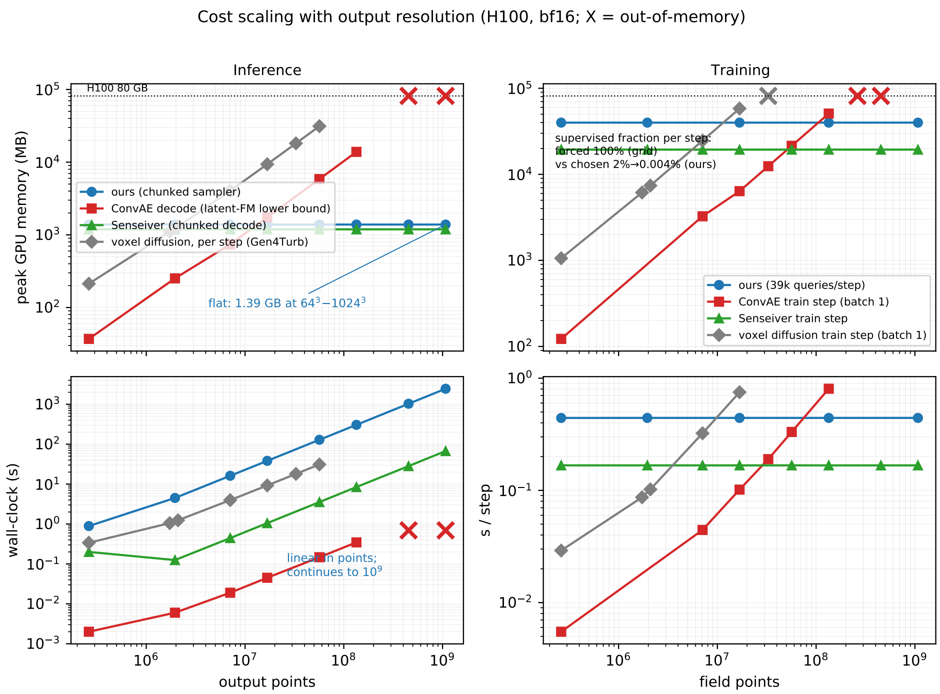

Figure 5: Full cost-scaling measurement: peak memory (top) and wall-clock (bottom) for inference and training. The wall-clock panels show the honest converse of the memory result—LOCUS's inference time is linear in the number of evaluation points and is the slowest in absolute terms at large resolution, while its training step time is flat.

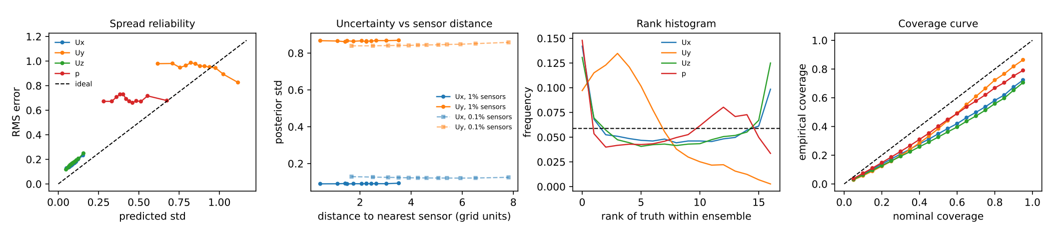

Figure 6: Pointwise uncertainty diagnostics on the held-out cube. Spread reliability: binned RMS error against predicted standard deviation, ideal on the diagonal; the curves are monotone but above it, the pointwise signature of underdispersion. Sensor distance: the profile is flat at 1% coverage and remains flat at 0.1%, where nearest-sensor distances reach eight cells—spread is set by channel identity and flow structure, not by distance to the nearest reading. Rank histogram: U-shaped for the observed channels (underdispersed) and sloped for U_y (biased, as expected for a channel no sensor constrains). Coverage curve: empirical against nominal central-interval coverage.

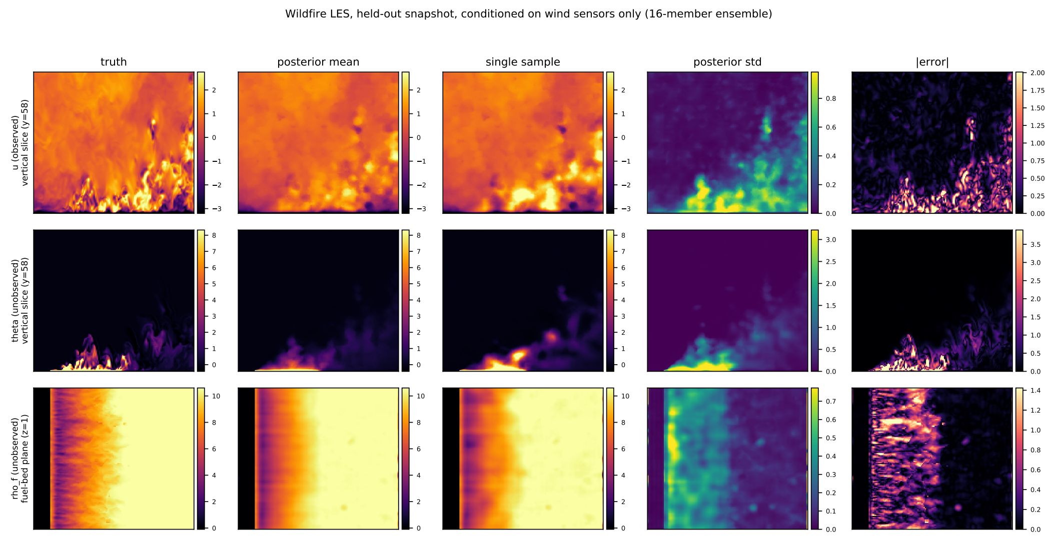

Figure 7: Wildfire LES, held-out snapshot, conditioned on wind sensors only. Wind and potential temperature are shown on a vertical slice through the plume; fuel density is shown on the fuel-bed plane, where its structure lives. Potential temperature and fuel density are never observed. From wind alone the model localizes the burn scar and its ragged front, and the predictive standard deviation concentrates on that front—the reconstruction is confident in the burned and unburned interiors and uncertain exactly where the front sits.

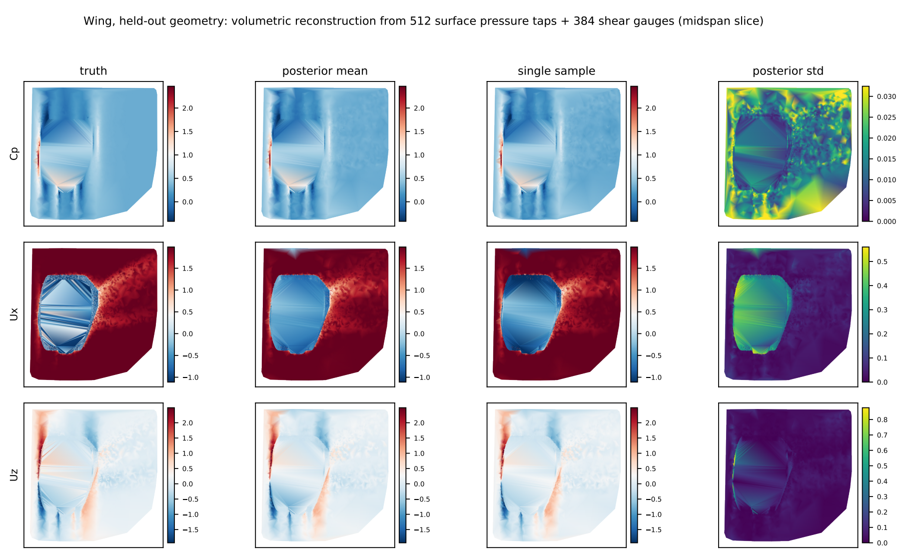

Figure 8: Wing, held-out geometry, midspan slice: volumetric reconstruction from surface pressure taps and shear gauges alone. The predictive standard deviation is largest in the boundary layer and wake—the regions surface data does not determine—and the standard deviation and error decay together away from the skin (Figure 9).

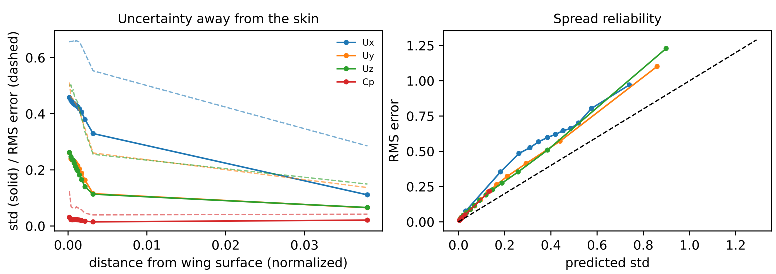

Figure 9: Wing uncertainty profiles. Left: predictive standard deviation (solid) and RMS error (dashed) against distance from the wing surface; both decay together, so the spread is informative about where the reconstruction is trustworthy. Right: spread reliability, again monotone and above the diagonal. Pointwise correlation between standard deviation and error is 0.77 on this case.